MTH 203 : Lecture 05

Partial Derivatives:

Recall Derivative for one Variable:

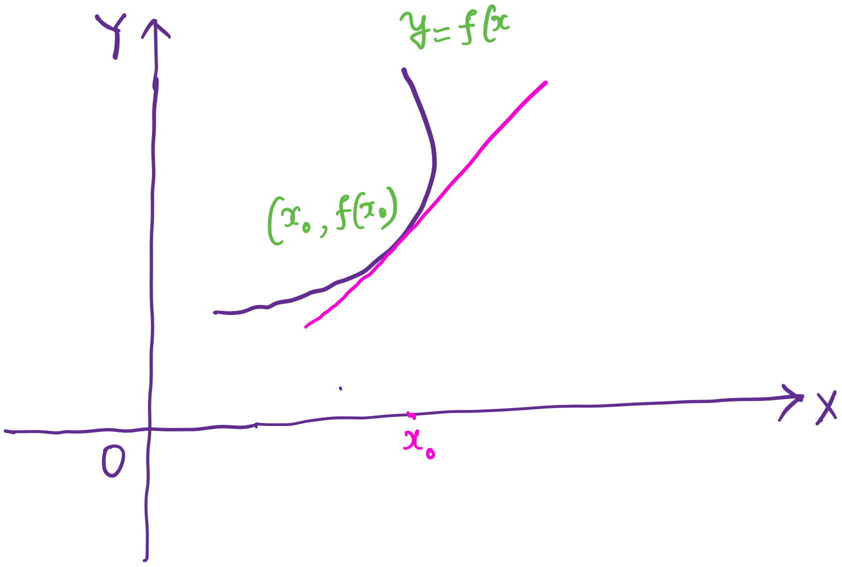
Let $D \subseteq \mathbb{R}$ and $f : D \longrightarrow \mathbb{R}$

f is called Differentiable (has a derivative at $x_0 \in D$) at $x_0 \in D$

if $\lim\limits_{h \to 0} \dfrac{f(x_0+h) - f(x_0)}{h}$ exists.

The value of the limit is called the derivative of f at x_0 and is denoted by $f'(x_0)$ or $\left.\dfrac{df}{dx}\right|_{x=x_0}$

Geometrical Interpretation:



$f'(x_0)$ is the slope of the tangent line to the curve $y = f(x)$ at the point $(x_0, f(x_0))$

Question: What is the analogous notion for two variables?

Partial Derivative:

Let $D \subseteq \mathbb{R}^2$, $f : D \to \mathbb{R}$ and $(x_0, y_0) \in D$
((x_0, y_0) is an interior point of D)
We say that f has a partial derivative with respect to x at the point (x_0, y_0) if

$$\lim_{h \to 0} \frac{f(x_0+h, y_0) - f(x_0, y_0)}{h} \text{ exists.}$$

The value of the limit is called the Partial Derivative of f with respect to x at the point (x_0, y_0) and is denoted by

$$\boxed{\left.\frac{\partial f}{\partial x}\right|_{(x_0, y_0)}} \quad \text{or} \quad \boxed{f_x(x_0, y_0)}$$

Similarly we say that f has a partial derivative with respect to y at the point (x_0, y_0) if

$$\lim_{h \to 0} \frac{f(x_0, y_0+h) - f(x_0, y_0)}{h} \text{ exists.}$$

The value of the limit is called the Partial Derivative of f with respect to y at the point (x_0, y_0) and is denoted by

$$\boxed{\left.\frac{\partial f}{\partial y}\right|_{(x_0, y_0)}} \quad \text{or} \quad \boxed{f_y(x_0, y_0)}$$

Examples: $f(x,y) = xy + 3x^2 + y$

compute $\frac{\partial f}{\partial x}$ at $(1,2)$

$\frac{\partial f}{\partial x} = y + 6x$, $\frac{\partial f}{\partial x}(1,2) = 2 + 6\times 1$

$= \boxed{8}$

Ex: $f(x,y) = x^3 - 3x^2y^3 + y^2$

$$\left.\begin{aligned}\frac{\partial f}{\partial x} &= 3x^2 - 6xy^3 \\ \frac{\partial f}{\partial y} &= -9x^2y^2 + 2y\end{aligned}\right\}$$

Ex: Using definition find f_x and f_y at $(1,2)$ when $f(x,y) = 1 - x + y - 3x^2y$.

$$f_x(1,2) = \lim_{h\to 0} \frac{f(1+h,2) - f(1,2)}{h}$$

$$= \lim_{h\to 0} \frac{[1-(1+h)+2-3(1+h)^2 2] - [1-1+2-3(1)^2 2]}{h}$$

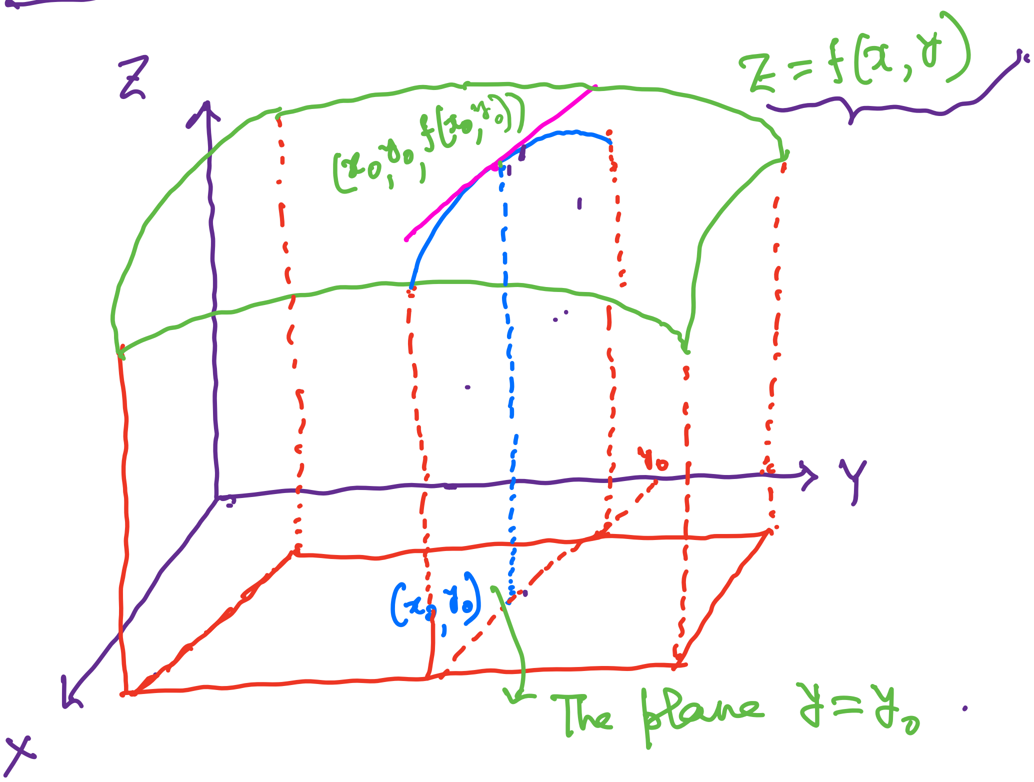
$$= \lim_{h\to} \frac{\cancel{1}-\cancel{1}-h+2-6(1+2h+h^2) - (-4)}{h} = \lim_{h\to 0} \frac{-13h-6h^2}{h}$$

$$= \lim_{h\to 0} \frac{\cancel{h}(-13-6h)}{\cancel{h}} = \boxed{-13}$$

$$f_y(1,2) = \lim_{h\to 0} \frac{f(1,2+h) - f(1,2)}{h} = \lim_{h\to 0} \frac{1-1+(2+h)-3(1)^2(2+h)-(-4)}{h}$$

$$= \lim_{h\to 0} \frac{2+h-6-3h+4}{h} = \lim_{h\to 0} \frac{-2h}{h} = \boxed{-2}$$

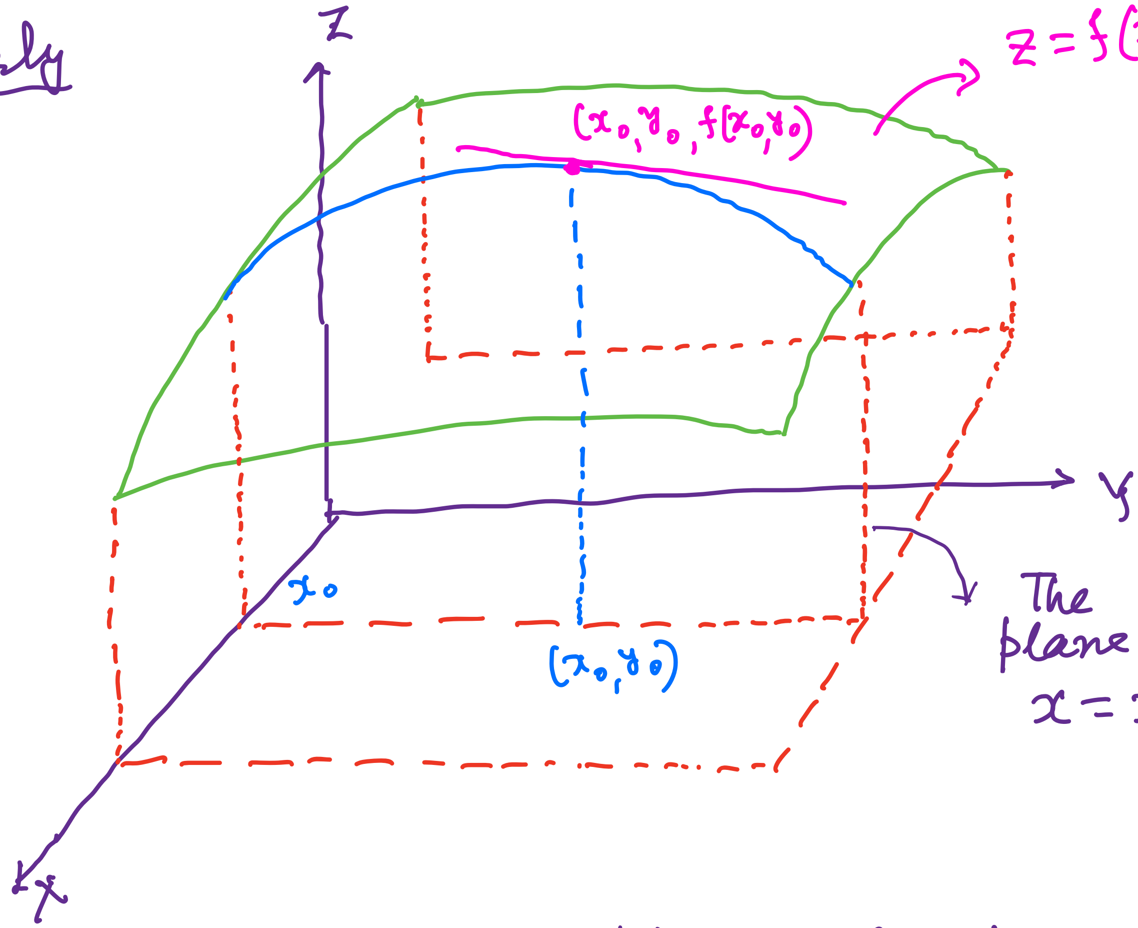
Geometrical Interpretation of Partial Derivatives for a function $z = f(x,y)$ of two variables



- The graph of $z = f(x,y)$ is a surface.
- $(x_0, y_0, f(x_0,y_0))$ is a point on the surface.
- We fix $y = y_0$: This is a plane.
- The intersection of the surface $z = f(x,y)$ with the plane $y = y_0$ is a curve $z = f(x, y_0)$ (contained in the plane $y = y_0$)
- Now $\left.\dfrac{\partial f}{\partial x}\right|_{(x_0,y_0)} \equiv f_x(x_0, y_0)$ is the slope of the tangent line to the curve $z = f(x, y_0)$ at $x = x_0$.

The surface $z = f(x, y)$

Similarly



The plane $x = x_0$

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0, y_0)} = f_y(x_0, y_0) = \text{slope of the tangent line to the curve } z = f(x_0, y) \text{ at } y = y_0$$

- Similarly we can define partial derivatives for a function of three variables $f : D \subseteq \mathbb{R}^3 \longrightarrow \mathbb{R}$

$$w = f(x, y, z)$$

Let $(x_0, y_0, z_0) \in D$ (an interior point of D)

If $\displaystyle\lim_{h \to 0} \frac{f(x_0+h, y_0, z_0) - f(x_0, y_0, z_0)}{h}$

exists, we call it $\left.\dfrac{\partial f}{\partial x}\right|_{(x_0,y_0,z_0)}$

or, $\left.\dfrac{\partial w}{\partial x}\right|_{(x_0,y_0,z_0)}$ or $f_x(x_0,y_0,z_0)$

Similarly we can define $\left.\dfrac{\partial f}{\partial y}\right|_{(x_0,y_0,z_0)}$ and $\left.\dfrac{\partial f}{\partial z}\right|_{(x_0,y_0,z_0)}$

Ex: Let $f : D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$

Let $(x_1^0, x_2^0, \ldots, x_n^0) \in D$ (an interior point of D)

Define the partial derivatives.

(There are n first order partial derivatives.)

Ex: $f(x,y) = \dfrac{x}{x^2+y^2}$

$$\frac{\partial f}{\partial x} = \frac{(x^2+y^2)\frac{\partial x}{\partial x} - x\frac{\partial (x^2+y^2)}{\partial x}}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2)(1) - x(2x)}{(x^2+y^2)^2} = \boxed{\frac{y^2-x^2}{(x^2+y^2)^2}}$$

$$\frac{\partial f}{\partial y} = x \frac{\partial \left\{(x^2+y^2)^{-1}\right\}}{\partial y} = x(-1)(x^2+y^2)^{-1-1}(2y)$$

$$= \boxed{\frac{-2xy}{(x^2+y^2)^2}}$$

Ex: $f(x,y) = x^y$

$$\frac{\partial f}{\partial x} = \boxed{y\,x^{y-1}} \qquad \frac{\partial f}{\partial y} = \boxed{x^y \ln x}$$

Ex: $f(x,y) = \log_y x = \dfrac{\ln x}{\ln y}$

$$\frac{\partial f}{\partial x} = \boxed{\frac{1}{x \ln y}}$$

$$\frac{\partial f}{\partial y} = \ln x\,(-1)(\ln y)^{-1-1}\,\frac{1}{y} = \boxed{-\frac{\ln x}{y(\ln y)^2}}$$

Ex: $f(x,y) = \int_x^y g(t)\,dt$ (g is a continuous function of t)

$= -\int_y^x g(t)\,dt$

$$\frac{\partial f}{\partial x} = \boxed{-g(x)}$$

$$\frac{\partial f}{\partial y} = \boxed{g(y)}$$

Ex: $f(x,y) = \sum_{n=0}^{\infty} (xy)^n$ where $|xy| < 1$

$= \frac{1}{1-xy}$

$$\frac{\partial f}{\partial x} = (-1)(1-xy)^{-1-1}(-y) = \boxed{\frac{y}{(1-xy)^2}}$$

$$\frac{\partial f}{\partial y} = (-1)(1-xy)^{-1-1}(-x) = \frac{x}{(1-xy)^2}$$

Ex: $f(x,y) = e^{xy}\ln y$

Then $\frac{\partial f}{\partial x} = (\ln y)\, e^{xy}\,(y) = \boxed{y \ln y\, e^{xy}}$

and $\frac{\partial f}{\partial y} = (\ln y)\, e^{xy}\,(x) + e^{xy}\,\frac{1}{y}$

$$= \boxed{e^{xy}\left(x \ln y + \frac{1}{y}\right)}$$

Ex: The plane $x = 1$ intersects the surface $z = 2x^2 + 3y^2$ in a curve at $(1, 3, 29)$

Find the slope of the tangent to the curve at $(1, 3, 29)$

$$\text{The slope} = \left.\frac{\partial z}{\partial y}\right|_{(1,3,29)} = 6y\Big|_{(1,3,29)}$$

$$= 6 \times 3 = \boxed{18}$$

Ex: Find the value of $\frac{\partial z}{\partial x}$ at $(1, 1, 1)$ if the equation $xy + z^3x - 2yz = 0$ defines z as a function of two independent variables x and y and partial derivative exists.

$$xy + z^3 x - 2yz = 0 \text{ (Implicit function)}$$

$$\frac{\partial(xy)}{\partial x} + \frac{\partial(z^3x)}{\partial x} - 2\frac{\partial(yz)}{\partial x} = 0$$

$$\Rightarrow y\frac{\partial x}{\partial x} + z^3\frac{\partial x}{\partial x} + x\,3z^2\frac{\partial z}{\partial x} - 2y\frac{\partial z}{\partial x} = 0$$

$$\Rightarrow y(1) + z^3(1) + 3xz^2\frac{\partial z}{\partial x} - 2x\frac{\partial z}{\partial x} = 0$$

$$\Rightarrow \frac{\partial z}{\partial x}\left(3xz^2 - 2y\right) = -y - z^3$$

$$\Rightarrow \frac{\partial z}{\partial x} = \frac{-y - z^3}{3xz^2 - 2y}$$

So, at $(1,1,1)$ $\quad \frac{\partial z}{\partial x} = \frac{-1 - 1^3}{3(1)(1)^2 - 2(1)} = \frac{-2}{3-2} = \boxed{-2}$

Ex: Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$

$$f(x,y) = \frac{x^2 y}{x^4 + y^2} \quad \text{for } (x,y) \neq (0,0)$$

$$= 0 \quad \text{for } (x,y) = (0,0)$$

Then $(0,0) \in \text{Domain}(f)$ and $f(0,0) = 0$.

Questions:

$$\left.\frac{\partial f}{\partial x}\right|_{(0,0)} = ? \qquad \left.\frac{\partial f}{\partial y}\right|_{(0,0)} = ?$$

$$\left.\frac{\partial f}{\partial x}\right|_{(0,0)} = \lim_{h\to 0} \frac{f(0+h,0) - f(0,0)}{h}$$

$$= \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{\frac{h^2 \times 0}{h^4+0^2} - 0}{h}$$

$$= \lim_{h\to 0} \frac{0}{h} = 0$$

$$\left.\frac{\partial f}{\partial y}\right|_{(0,0)} = \lim_{h\to 0} \frac{f(0,0+h) - f(0,0)}{h}$$

$$= \lim_{h\to 0} \frac{f(0,h) - f(0,0)}{h}$$

$$= \lim_{h\to 0} \frac{\frac{0^2 h}{0^4+h^2} - 0}{h} = \lim_{h\to 0} \frac{0}{h} = 0$$

So, $\left.\frac{\partial f}{\partial x}\right|_{(0,0)} = 0 \qquad \left.\frac{\partial f}{\partial y}\right|_{(0,0)} = 0$

<u>Question:</u> Is f continuous at $(0,0)$?

We can say that $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ doesn't exist (Use two path test with $y = Kx^2$)

Note: A function $f(x,y)$ may have partial derivatives f_x and f_y at a point but may be discontinuous there.

Ex: $$f(x,y) = \begin{cases} 0 & \text{if } xy \neq 0 \\ 1 & \text{if } xy = 0 \end{cases}$$

$$\lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = 0 \Rightarrow f_x(0,0) = 0$$

$$\lim_{h\to 0} \frac{f(0,h) - f(0,0)}{h} = 0 \Rightarrow f_y(0,0) = 0$$

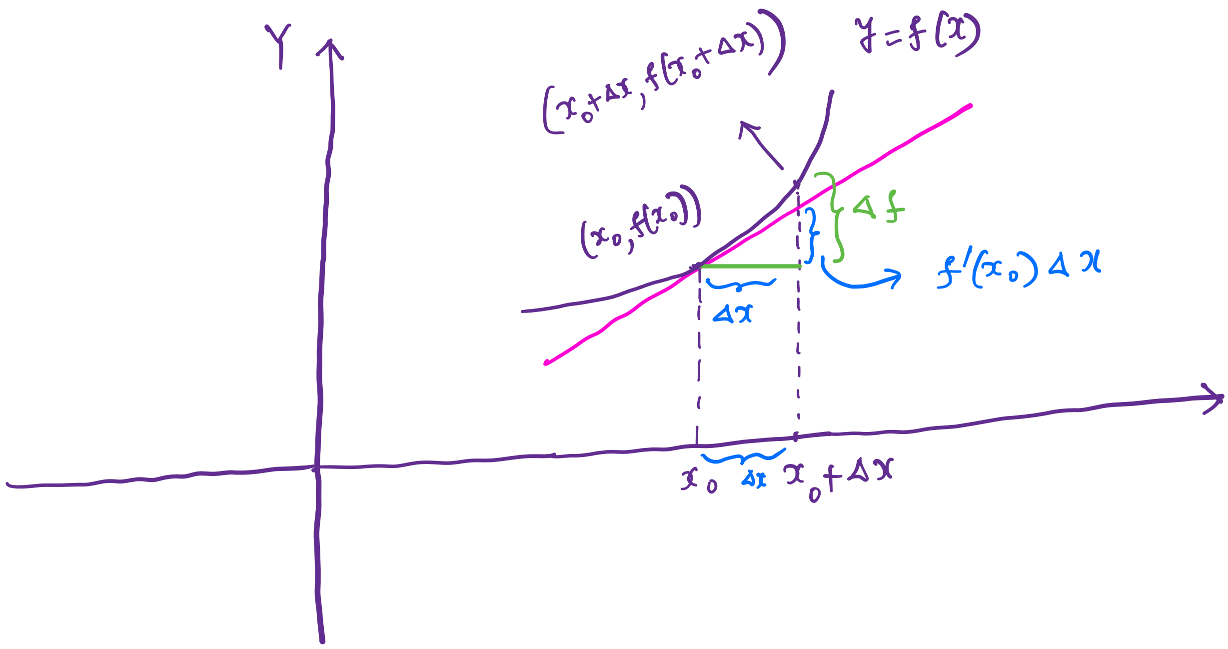
But $\lim\limits_{\substack{(x,y)\to(0,0)\\ \text{along } y=x \\ (x\neq 0)}} f(x,y) = 0$

and $\lim\limits_{\substack{(x,y)\to(0,0)\\ y=0 \\ (\text{i.e. along } x\text{-axis})}} f(x,y) = 1$

So, $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ doesnot exist.

Hence f is discontinuous at $(0,0)$.

Let us look at a function of one variable:



Let $\Delta f = f(x_0+\Delta x) - f(x_0)$

and f is differentiable at x_0

Then $\Delta f - f'(x_0)\Delta x = \epsilon \Delta x$

where $\epsilon \to 0$ as $\Delta x \to 0$

Proof:

$$\Delta f - f'(x_0)\Delta x$$

$$= f(x_0+\Delta x) - f(x_0) - f'(x_0)\Delta x$$

$$= \frac{f(x_0+\Delta x) - f(x_0)}{\Delta x}\Delta x - f'(x_0)\Delta x$$

$$= \underbrace{\left[\frac{f(x_0+\Delta x) - f(x_0)}{\Delta x} - f'(x_0)\right]}_{\epsilon \text{ (say)}}\Delta x$$

$$= \epsilon \Delta x$$

Now if $f(x)$ has derivative at x_0
(i.e. differentiable at x_0)

then $\epsilon \longrightarrow 0$ as $\Delta x \longrightarrow 0$

Thus we reach the conclusion that

$$\boxed{\begin{array}{l} \Delta f - f'(x_0)\Delta x = \epsilon \Delta x \\ \text{where } \epsilon \longrightarrow 0 \text{ as } \Delta x \longrightarrow 0 \end{array}}$$

(QED)

Conversely

Suppose $\exists\ a \in \mathbb{R}$ such that

$\Delta f - a\Delta x = f(x_0 + \Delta x) - f(x_0) - a\Delta x = \epsilon_1 \Delta x$

where $\epsilon_1 \longrightarrow 0$ as $\Delta x \longrightarrow 0$

Then f is differentiable at x_0 and $f'(x_0) = a$

Proof:

$$\Delta f - a\Delta x = f(x_0+\Delta x) - f(x_0) - a\Delta x = \epsilon_1 \Delta x$$

$$\Rightarrow \frac{f(x_0+\Delta x) - f(x_0)}{\Delta x} - a = \epsilon_1$$

$$\Longrightarrow \lim_{\Delta x \to 0} \frac{f(x_0+\Delta x) - f(x_0)}{\Delta x} = a + \lim_{\Delta x \to 0} \epsilon_1$$

$$\Longrightarrow \lim_{\Delta x \to 0} \frac{f(x_0+\Delta x) - f(x_0)}{\Delta x} = a$$

So, f is differentiable at x_0 and $f'(x_0) = a$

Thus we see that

f is differentiable at x_0

$\Leftrightarrow \Delta f - f'(x_0)\Delta x = \epsilon \Delta x$

where $\epsilon \longrightarrow 0$ as $\Delta x \longrightarrow 0$

Differentiability (of a function of two variable)

Let $Z = f(x,y)$

f is called differentiable at (x_0, y_0) if $\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)}$ and $\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)}$ exist

and $\Delta f = f(x_0+\Delta x, y_0+\Delta y) - f(x_0,y_0)$

$$= \left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} \Delta x + \left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} \Delta y$$

$$+ \epsilon_1 \Delta x + \epsilon_2 \Delta y$$

where $\epsilon_1, \epsilon_2 \longrightarrow 0$ as $\Delta x, \Delta y \longrightarrow 0$

Theorem: If for a function $f(x,y)$, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist and continuous in an open region $D \subseteq \mathbb{R}^2$, then f is differentiable at every point of D.

<u>Theorem:</u> If $f(x,y)$ is differentiable at (x_0,y_0), then f is continuous at (x_0,y_0).

<u>Ex:</u>
$$f(x,y) = \begin{cases} 0 & \text{for } xy \neq 0 \\ 1 & \text{for } xy = 0 \end{cases}$$

Is f differentiable at $(0,0) \in D$?

<u>Ans:</u> f can't be differentiable at $(0,0)$ because we have already seen that f is not continuous at $(0,0)$.

<u>Note:</u> We will revisit the concept of Differentiability using the concept of Linear Transformation.